

Autoparty: Machine Learning-Guided Visual Inspection of Molecular Docking Results

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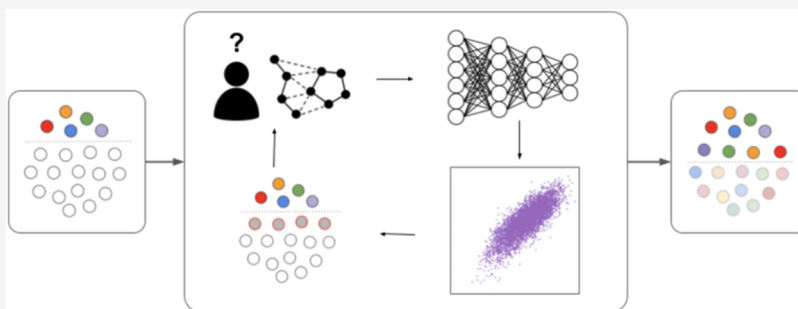
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ABSTRACT: Human inspection of potential drug compounds is crucial in the virtual drug screening pipeline. However, there is a pressing need to accelerate this process, as the number of molecules humans can realistically examine is extremely limited relative to the scale of virtual screens. Furthermore, computational medicinal chemists can evaluate different poses inconsistently, and there is no standard way of recording annotations. We propose Autoparty, a containerized tool to address these challenges. Autoparty leverages on-premises active learning for drug discovery to facilitate human-in-the-loop training of models that extrapolate human intuition. We leverage multiple uncertainty quantification metrics to query the user with informative examples for model training, limiting the number of human expert training labels. The collected annotations populate a persistent and exportable local database for broad downstream uses. Incorporating Autoparty resulted in a 40% increase in hit rate over shape similarity alone among 193 experimentally tested compounds in a real-world case study.

INTRODUCTION

Computer-aided drug design (CADD) techniques are often used in the early stages of drug discovery to identify lead molecules with high predicted binding affinity against a specific target.^{1–3} Virtual screening (VS) programs such as DOCK^{4,5} and Glide^{6,7} computationally dock and score libraries of compounds, producing ranked lists of molecules and their predicted protein–ligand structures. The top-scoring molecules are typically clustered by chemical structure and often filtered against known binders to discover novel chemotypes. Additional filtering steps based on interaction fingerprints or alternative scoring methods may be applied during this step as well. The remaining best-scoring cluster heads are often visually inspected (“hit picked”) by expert computational medicinal chemists to evaluate the quality of the generated poses before selecting final compounds for experimental testing (Figure 1a).^{8,9}

False positive molecules, which rank highly by the scoring function but fail to bind when tested experimentally, are a primary difficulty in this VS pipeline. Incorrect binding pose¹⁰ or limited scoring function accuracy arising from biophysical abstractions and difficulty weighting scoring terms^{11,12} underlie typical docking failures. Recent binding affinity prediction

methods, including ABFEP,¹³ have been used to rerank compounds^{13,14} and identify docking artifacts¹¹ but remain limited in application due to high time and compute costs. While machine learning approaches can achieve high accuracy on experimental affinity data sets, they suffer from a lack of negative data and display variable performance across protein targets and chemical series.¹⁵

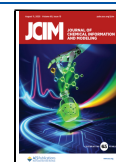
Consequently, computational medicinal chemists commonly manually inspect the top-scoring compounds from virtual screens to minimize false positives and improve potency. A review of drug discovery protocols¹⁶ found visual inspection of docked poses in 50% of the 250 publications surveyed, often as the final step in compound prioritization before experimental testing. Multiple scientists often collaborate in visual inspection, working toward a consensus during “Hit Picking Parties.”³

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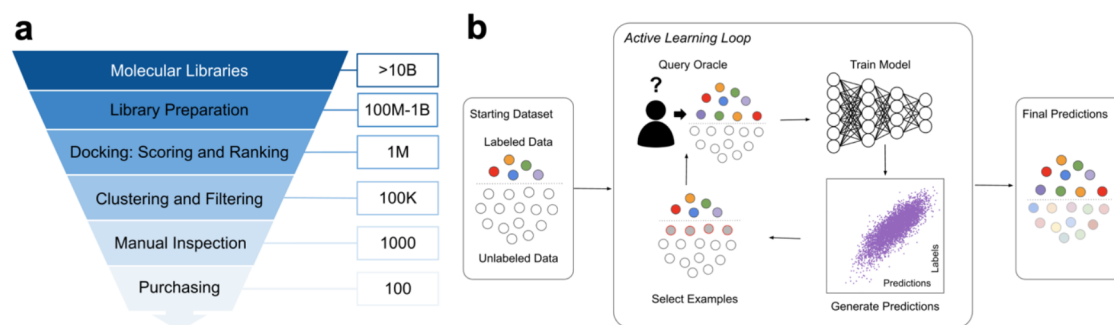


Figure 1. Overview of the virtual screening pipeline and active learning. (a) The drug-discovery “funnel” shows standard screening steps with the number of molecules remaining (approximate), motivating the need for tools to ensure promising molecules progress to experimental testing. (b) Human-in-the-loop active learning training paradigm. The starting data set lacks many true labels. The oracle (human) initially labels a random subset for model training. The initial model predicts labels and their uncertainty for the unannotated data set. The oracle annotates selected examples. This process repeats until it reaches a label quota or the desired accuracy.

Large-scale virtual screening campaigns incorporating this inspection step achieve hit rates between 10–40%^{17–24} and exceed 60% in some rare cases.^{25,26} It has been suggested that any sufficiently experienced human could outperform existing scoring functions for compound selection.²⁷ Indeed, studies have found that molecules chosen by expert computational medicinal chemists achieve greater potencies than those selected by scoring functions alone.^{17,24}

However, there are limits to the number of poses a single scientist or team can realistically examine. Pragmatism restricts manual inspection to the top screening results, thus this approach does not scale to the remaining well-ranked molecules, a limitation that generates several problems. As the size of available molecular libraries increases into the order of billions and beyond,²⁸ the number of molecules with good docking scores likewise expands.²⁹ Recent reports of a hit rate plateau suggest there may be a significant number of false negatives further down in the screening results,^{17,26} an issue that remains largely unexplored due to the bottleneck of visual inspection. Finally, expert assignments may vary or reflect individual areas of expertise; despite efforts to codify rules for visual inspection,^{9,30} there remains no high-quality database of expert annotations for learners to use as a reference. This problem is further complicated by the fact that even expert medicinal chemists often disagree with one another.³¹ A tool that could emulate human labels would better scale with chemical library growth, minimize contextual variability, provide a reliable reference for scientist training, and enable progressive improvements of filtering strategies.

One approach to creating such a tool is active learning (AL).^{32,33} Active learning is a semi-supervised machine learning technique for training models with minimal labeled data, particularly when labels are expensive to obtain. In AL, the model is first trained on a random subset of the data. The model then predicts labels for and selects examples from the remaining unlabeled data for submission to the “oracle” to obtain true labels. These examples and labels are then added to the training set for the next iteration (Figure 1b). Intuitively, this approach can identify the most informative examples at each iteration, constructing an accurate model from minimal data. The oracle is often a human expert, motivating the term human-in-the-loop training. In virtual screening, AL has been used to select regions of chemical space to dock from large libraries. The oracle is the docking program, and the model learns to select compounds to build and dock. Yang et al.³⁴ demonstrated the success of this

approach across multiple receptors both retrospectively and prospectively for UCSF DOCK. Graff et al.³⁵ applied AL to screens from small libraries (Enamine 10K) to ultralarge libraries (>100M). They showed improved recovery of top hits over random search for Glide and DOCK scores. Thompson et al.³⁶ explored various AL parameters, including acquisition function, number of molecules sampled at each iteration, ML architecture, and initial sampling strategy for predicting the output of free energy perturbation (FEP) calculations. Their best-performing models were trained on only 6% of the total data set and successfully recovered 75% of the top 100 molecules, substantially decreasing the required number of calculations. These studies support AL as an efficient paradigm for learning complex scoring functions.

In this application note, we introduce Autoparty, a tool for AI-assisted accelerated hit picking on molecular docking results. Autoparty is a browser-based tool for visualizing the protein–ligand interface and intermolecular interactions. The user assigns grades to each molecule, which are stored in a database and available for export. Autoparty uses active learning with human-in-the-loop training to display the most useful poses for the user to annotate. This trained model can then be applied to more docked compounds, acting as an additional filter during hit picking and facilitating the recovery of lower-ranked but promising false negatives. We hope this tool will bring together ML advancements with medicinal chemical expertise to facilitate pose evaluation and recovery of otherwise hidden promising compounds.

RESULTS

Autoparty: A Tool for Automated Human-in-the-Loop Molecule Inspection. Autoparty is a Python-based containerized application developed to assist scientists in analyzing the results of virtual docking screens using active deep learning. The user uploads the protein target and docked ligands to visualize their structure and relevant intermolecular interactions. The user can then grade these molecules. These assigned grades are saved to a SQL database to maintain a robust record of structures and human annotations. During this human-in-the-loop training, a machine learning (ML) model is trained to predict annotations for new molecules and determine which molecules are the most informative for the user to inspect manually. This process continues until the user finishes grading molecules, after which the full list of grades and model predictions can be exported for further review (Figure 2). In

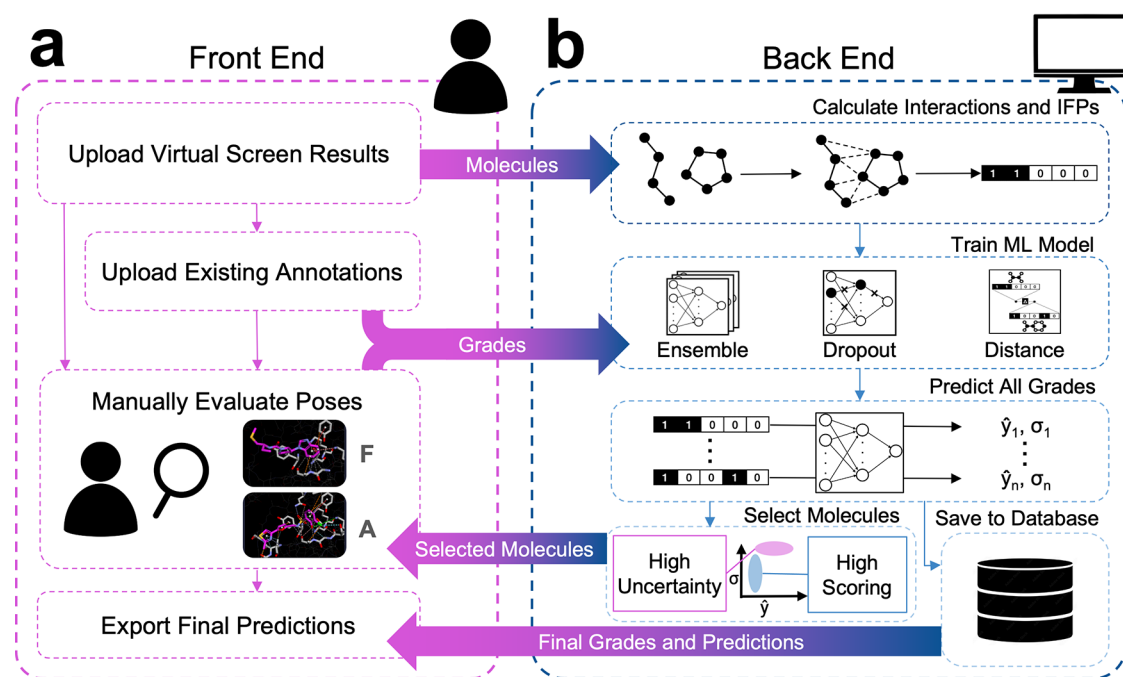


Figure 2. Schematic of Autoparty workflow and user interaction. (a) The user interacts with Autoparty’s “front end” functions. These include uploading the initial screening results, importing optional pre-existing annotations, and grading new molecules. The user can also export all existing grades and predictions from the database. Displayed molecules are ZINC0D000000Z8ck (F) and ZINC0F000005aGQZ (A) docked into AmpC β -lactamase. (b) “Back end” Autoparty functions automatically assist with hit picking. Autoparty calculates each protein–ligand complex’s interactions and LUNA interaction fingerprints (IFPs) upon upload. Autoparty trains an ML model with the architecture and confidence estimation procedure selected by the user. Available options include ensemble, in which the variance between individual models is used as a measure of confidence; dropout, which repredicts the same example multiple times while randomly “dropping” neural network weights to calculate a distribution of scores; and distance, which uses the topological similarity of new examples to those used in the training set (full options detailed in [Supporting Methods](#)). The back end also saves the user-provided grades, trains models to predict human labels, and orders compounds by the acquisition function.

addition, the trained models can be extracted and applied to new molecules not present in the initially uploaded screen. In the next few paragraphs, we describe each of these steps and their associated configuration options in greater detail. A full list of Autoparty settings with their defaults is provided in [Supporting Table 1](#).

Interaction Calculation and Representation Generation. When a screen is uploaded to Autoparty, the LUNA toolkit³⁷ calculates intermolecular interactions between all input molecules and the protein structure. This process comprises two steps: First, the biochemical properties of the ligand and protein atoms at the interface are determined using openbabel³⁸ or RDKit.³⁹ This includes the determination of atomic groups, such as aromatic rings, that participate as a single interacting group. Groups are visualized in Autoparty as white spheres at the group’s center of mass. Second, interactions are defined as pairs of atoms or atomic groups in proximity that meet certain geometric and chemical criteria. Common interaction types include van der Waals interactions, hydrogen bonds, hydrophobic interactions, halogen bonds, and others. These interactions are shown on the complex as dashed cylinders. All available interaction types and associated display colors are provided in [Supporting Table 2](#).

We used interaction fingerprints (IFPs) as the backend ML model’s input. In contrast to molecular fingerprints that encode the chemical topology of molecules alone, IFPs capture the geometric and biophysical properties of the full protein–ligand interface. They have been used to predict docking scores and binding affinities^{37,40} and help address activity cliffs in ligand-based machine learning scoring functions.⁴¹ Fingerprint

representations perform comparably to more complex message-passing neural networks at small data set sizes.³⁵

The ligand and interactions are divided into multiple nested spherical shells centered on each ligand atom to convert the calculated interaction graph into a fixed-size vector representation of the protein–ligand interface. The first shell at radius 0 contains the initial atomic identifiers for each ligand atom, including their “invariant” properties. These invariants can be explicit, including the number of non-hydrogen neighbors, total bond order, atomic number, atomic mass, atomic charges, and number of attached hydrogens, or functional/pharmacophoric (aromatic, acceptor, donor, hydrophobic, etc.). EIFP and FIFP describe these explicit or functional interaction fingerprint approaches, respectively. Progressively larger shells centered on the atoms or atom-group centers of mass contain the interactions within their boundaries. Each shell is passed through a hash function to generate a series of “on” indices, which are collected into the final fingerprint following the extended-connectivity fingerprint protocol introduced by Rogers and Hahn.⁴² This one-time calculation occurs before the molecule is saved to the database, and the user can export the calculated IFPs. By default, Autoparty uses EIFP count fingerprints calculated with 2 shells of radius approximately 6 Å and folded to a length of 4096, per the optimal settings for DOCK score prediction in previous studies.³⁷ Fassio et al.³⁷ describe this fingerprint generation protocol and its initial chemical features in depth.

Model Architecture and Training. In standard AL workflows, models provide a predicted label and a measure of its uncertainty quantification (UQ). These outputs are used in

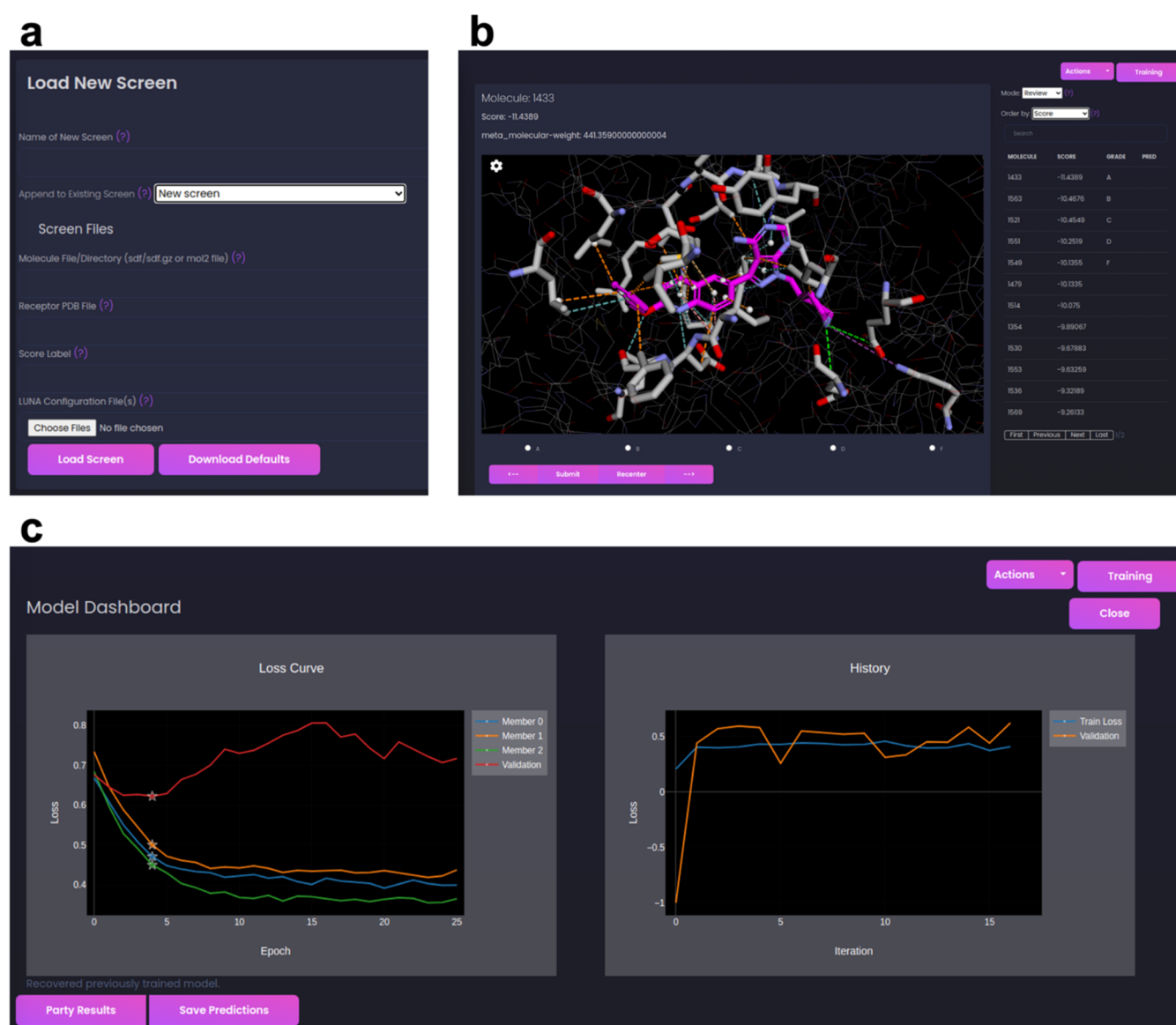


Figure 3. Autoparty web interface. (a) The screen upload interface. In addition to the required protein and separate molecules file, the user can specify an attribute to sort molecules for initial annotation. The user may also provide specific LUNA configurations for interaction calculation. (b) Hit picking interface for human-in-the-loop AL training. The current molecule appears with its calculated protein site interactions (calcium-dependent protein kinase CDPK1, PDB ID: 3T3U⁵³) above the potential grades to be assigned. The top right shows the current mode (Annotate vs. Review) and sorting method (Score, Uncertainty, Disagreement, Random) along with drop-downs to other Actions (e.g., screen data set options) and the model Training panel. (c) Model Dashboard showing training curves (left) and model accuracy over training iteration (right).

various acquisition strategies, such as greedy and least-confidence sampling, to determine the examples to include in the next AL iteration. The performance of UQ methods in molecular property prediction strongly depends on the data set itself.^{43,44} By default, Autoparty trains a committee of three unique, independent neural networks, each consisting of an input layer and two hidden layers of 1024 neurons. The variance across predictions from individual committee members defines the degree of prediction uncertainty.⁴⁵ The number of unique possible grades provided by the user determines the output layer size. Each ensemble member learns from initial unique training data, sampled with replacement from the entire set of available graded molecules. This approach synthetically diversifies the individual models while limiting overhead, as the training time for each component model remains short given the data set size.

Additionally, ensemble UQ provided high accuracy and good calibration in our testing (Supporting Figure 1 and Table 3).

Due to the variability of UQ methods across data sets, Autoparty includes other uncertainty estimation architectures as options for the user. Specifically, Autoparty allows for dropout-based uncertainty⁴⁶ in which a single model repeatedly predicts a label for a single example with random dropout. Similar to ensemble methods, the final prediction is the average of multiple predictions, and the uncertainty is their variance. Autoparty also includes the option of distance-based uncertainty, where the distance to the closest neighbors in the training set is a proxy for uncertainty.^{47,48}

Unlike typical classification labels, grades are ordinal, meaning that the output classes have an ordered relationship; A is closer to B than F. For ideal training, it is necessary to include this property in the formulation of output classes and the model's

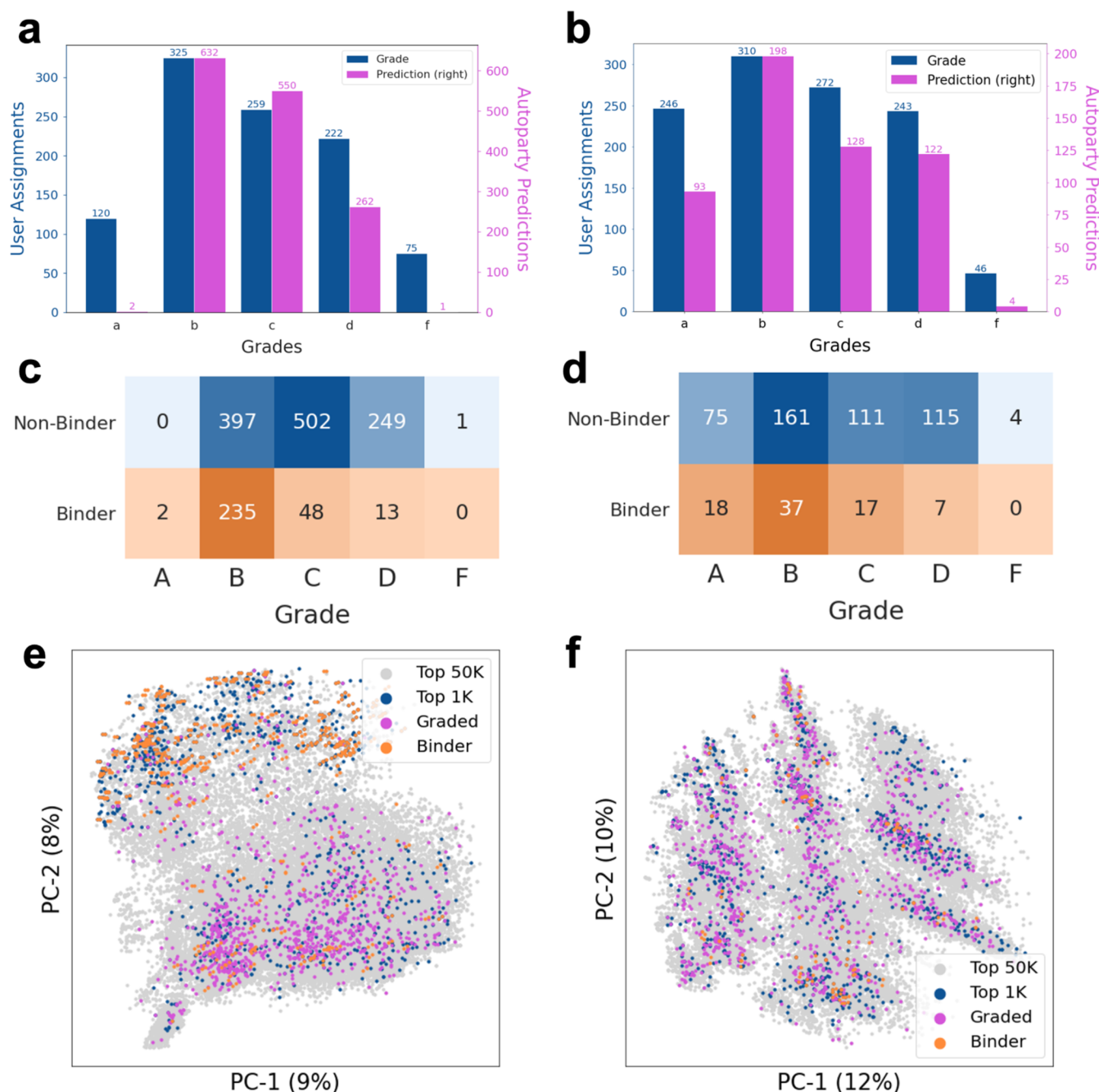


Figure 4. Retrospective Autoparty use-case study at UCSF. Counts of user-assigned grades (blue) and Autoparty-assigned grades for experimentally tested molecules (pink) for (a) AmpC and (b) D4. Autoparty grade predictions for all experimentally tested molecules against (c) AmpC and (d) D4. Principal component visualization of the chemical space covered by all molecules in the hit picking session (gray), those graded by the user (pink), the top-scoring molecules (blue), and the experimentally confirmed binders (orange) for (e) AmpC and (f) D4. Binders (orange) and their structurally similar molecules were proactively excluded from Autoparty training.

loss function. Some approaches treat ordinal labels as a regression task and bin the predictions into different classes, though this requires additional tuning of these cutoff values. We used a cumulative one-encoding strategy with labels as vectors of 1s followed by 0s. The count of 1s indicated the value of the grade (Additional detail provided in [Supporting Methods](#)). We trained models with binary cross-entropy loss. On a small data set of MM/GBSA scores of complexes from a virtual screen against the Dopamine D4 receptor, models using ordinal encoding labels achieved lower root-mean-square error between predicted and true classes compared to nominal (unordered)

encodings ([Supporting Figure 2](#)). Training with ordinal encodings also resulted in a lower number of false positive “A” predictions, a desirable quality for filtering out molecules during hit picking. In contrast, nominal grade encodings yielded overprediction of A and F grades and lost intermediate predictions. Autoparty still supports nominal class labels if requested, in which case the output classes are conventionally one-hot encoded.

The acquisition function is a critical AL design choice. The function defines the criteria for selecting examples at each oracle iteration. Historically, AL frameworks have prioritized examples

Table 1. Hit rates and molecule counts for experimentally validated molecules chosen with various selection criteria^a

	All	VI	DS	AP	VI&AP	DS&AP	~DS	~DS&AP
AmpC	20.6 (298/1447)	21.7 (149/687)	21.0 (251/1196)	37.4 (237/634)	33.8 (128/379)	36.5 (194/532)	2.1 (2/97)	0 (0/2)
D4	14.5 (79/545)	25.4 (31/122)	22.8 (26/114)	18.9 (55/291)	23.8 (25/105)	21.1 (16/76)	7.6 (25/328)	22.2 (6/27)

^aVI: Visual inspection, DS: Below dock score of worst-scoring molecule visually inspected during screen (AmpC: −60.0, D4: −63.68). AP: Autoparty-predicted AIB grade. ~DS: Above dock score of worst-scoring visually inspected molecule. ~DS&AP: Above dock score cutoff and predicted A by Autoparty. Logical operators denote Boolean set operations (&: And, ~: Not). All fractions display true binders over selected molecules. Best hit rate per row in **bold**, second best *italicized*.

with the highest calculated uncertainty as being maximally informative for the model.^{32,49,50} Studies using AL for virtual screening^{34–36} have also used greedy acquisition strategies that select the examples (molecules) with the best-predicted score regardless of uncertainty. This approach successfully found the true top-scoring molecules from large libraries of compounds, a more useful metric for their purposes than model accuracy alone. Similarly, Gusev et al.⁵¹ combined similarity clustering with greedy cluster-head selection to train a relative binding free energy prediction model. By contrast, Thompson et al.⁵² found that the number of top compounds recovered was surprisingly indifferent to acquisition strategy, although all strategies outperformed random acquisition. We provide greedy, maximum uncertainty, and random orderings as options to the user, with maximum uncertainty as the default.

Throwing a Hit Picking Party.

1. *Uploading a screen.* Autoparty's first step is uploading the result of a virtual docking screen (Figures 2a and 3). The user provides protein and docked molecule files. Additionally, the user indicates a property within the initial molecules file, such as docking score, to sort the compounds for annotation before any active learning. Autoparty reads molecules and their three-dimensional coordinates from the provided file and calculates intermolecular interactions and IFPs in batches. Hit picking begins as soon as the interactions for the first molecule are ready.
2. *Getting the party started.* The hit picking session can begin once the screen has started uploading and calculating interactions. The user either starts a new party by selecting the uploaded screen data set or resumes an existing party to recover existing grades, models, and predictions from a prior session. Supporting Table 1 provides all available settings and associated options.
3. *Annotating molecules.* At the start, Autoparty sorts molecules by the label provided by the user during upload (Figure 3). The full searchable list of molecules appears on the right. The available grades appear at the bottom of the screen. "Annotation" mode hides molecules with previously assigned grades, displaying only new molecules sorted by the selected acquisition function once the initial model has been trained. In "Review" mode (e.g., Figure 3b), the user can view and edit their annotated molecules, offering the opportunity for regrading. In Review mode, an additional "Disagreement" sorting method shows molecules with high-certainty predicted grades that differ from the user's previously assigned annotation for potential reexamination.
4. *Uploading existing annotations (optional).* Autoparty supports importing previous molecule screen annotations. Autoparty can update its internal database from a user-provided CSV file containing molecule names and previously assigned grades. These new grades count

toward the threshold for the number of total grades (annotations) to begin model training.

5. *Training a model.* After collecting a minimum number of annotations (default 100), Autoparty begins to train a model for the human-in-the-loop AL process. On the back end, the codebase extracts the fingerprints and corresponding annotations from the current party from the database to use as a training set and trains a model to predict these labels (Figure 2b). The tool then applies the trained model to all unannotated molecules, predicting a) a grade for that molecule and b) the model's confidence in that predicted grade. Upon completion, Autoparty reorders the molecules based on the acquisition function. This process continues until stopped by the user. Autoparty updates the models' loss curves and history in real time, displaying them in the Training dashboard.

UCSF Retrospective Case Studies: AmpC β -lactamase and Dopamine D4 Receptor. At UCSF, we evaluated Autoparty's ability to identify experimental hits in the top 500,000 poses from large-scale AmpC β -lactamase²⁴ and Dopamine D4¹⁷ receptor screens from lsd.docking.org,⁵⁴ a database of molecular docking scores, poses, and in vitro experimental data. Usefully, these data sets contain experimentally tested compounds sampled from a large range of dock score ranks in addition to top-scoring compounds. To create a meaningful challenge, we removed all experimentally tested compounds and their structural neighbors, even top-scoring ones, from Autoparty's training. We clustered all molecules, removed clusters containing experimentally tested compounds, and uploaded the remaining cluster heads to Autoparty for grading (Supporting Methods). For AmpC, we graded molecules on their ability to engage the oxyanion hole, overall fit, and additional interactions with the protein. For D4, we graded molecules on their ability to fill the pocket formed by helices 5 and 6 and their interactions with D115. For both screens, we graded approximately 100 top-scoring molecules at a rate of roughly 500 molecules/hour, trained an initial Autoparty model, and graded an additional 1000 molecules ranked by model uncertainty. We then applied the final models trained on these annotations to the corresponding experimentally tested molecules, consisting of 1447 AmpC and 545 D4 compounds. The classification of binders/nonbinders was taken from previous studies (SI Data Sets 1–4).^{17,24}

Autoparty predicted few A and F grades for experimentally tested AmpC campaign molecules, comprising 0.13% and 0.07% of the predictions, respectively. (Figure 4a). Using Autoparty-predicted grades of A or B as an additional filter on compounds selected through Visual Inspection (VI) increased the hit rate from 21.7 to 33.8% (Table 1). When evaluated on all experimentally tested molecules, not just those chosen manually, Autoparty-selected compounds achieved a hit rate of 37.4%, higher than all other selection methods evaluated (Figure 4c and Table 1). Strikingly, the molecules inspected during Autoparty

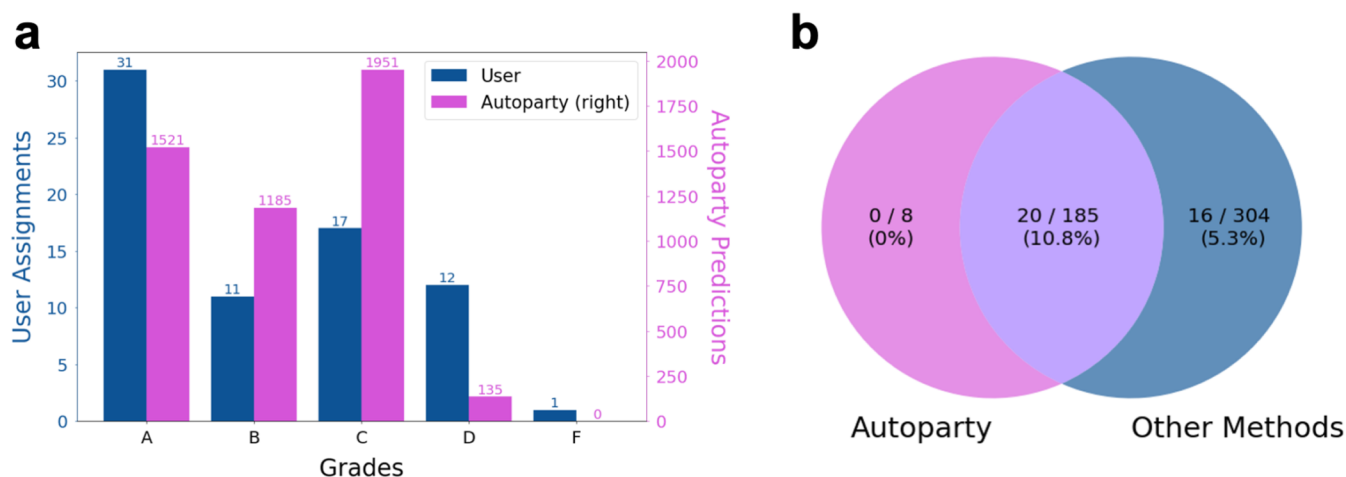


Figure 5. Autoparty prospective use-case study. (a) Counts of user-assigned (blue) and Autoparty-assigned (pink) grades. The two distributions are similar, though Autoparty assigns a higher proportion of compounds to C, likely corresponding with IFPs dissimilar to those that received annotations. Notably, Autoparty assigns no F grades, indicating that the model was unable to learn the pattern of human-assigned grades from a single annotation. The proportion of A-assigned grades was similar between the two methods, representing 43% of human annotations and 32% of predicted grades. (b) Venn diagram of compounds and hit rates selected by Autoparty vs alternative methods. The hit rate of molecules chosen by both methods had a hit rate exceeding either individual selection method.

Table 2. Comparison of hit rates with approaches based on user (visual) assignment, Autoparty assignments, and other methods (See Supporting Methods)^a

	AP	OM	AP&OM	AP&~OM	~AP&OM	VI&OM
# Tested	193	489	185	8	304	4
30 μ M	20 (10.4)	36 (7.4)	20 (10.8)	0 (0)	16 (5.3)	0 (0)
10 μ M	7 (3.6)	7 (1.4)	7 (3.8)	0 (0)	0 (0)	0 (0)
1 μ M	1 (0.52)	1 (0.20)	1 (0.54)	0 (0)	0 (0)	0 (0)

^aVI: Visual Inspection, AP: Autoparty, OM: Other methods. Logical operators denote Boolean set operations (&: And, ~: Not). Reported values are the number of compounds with IC₅₀ below the displayed cutoff followed by the hit rate (%). Best hit rate per row in **bold**, second best *italicized*.

hit picking sampled a broader region of chemical space than the top-ranking molecules by DOCK score and included regions containing confirmed but lower-ranked hits (Figure 4e).

Compared to AmpC, Autoparty predicted a greater proportion of A grades for D4 (17.1%) but few F grades (0.7%, Figure 4b). Unlike the AmpC campaign, using Autoparty grades of A or B as an additional filter on manually picked compounds had little effect on the hit rate (25.4% VI vs 23.8% VI&AP, Table 1). Relying solely on Autoparty grades of A or B resulted in a slightly worse hit rate of 19% (Figure 4d). Unlike AmpC, Autoparty's D4 campaign chemical space exploration tracked with the top-scoring molecules (Figure 4f). The two campaigns had different docked library sizes, with the top 500,000 poses making up the top 0.03% of the AmpC campaign (1.7 billion molecules) versus 0.3% of D4 (170 million molecules). Intuitively, the exploration patterns agreed with the human grader's assessment of likely AmpC ligands favoring more specific chemotypes than would the putative D4 ligands. Interestingly, on poorly scoring compounds outside of the region of visual inspection for this screen, Autoparty-A compounds achieved a hit rate of 22%, compared to 7.6% overall for these molecules (~DS&AP vs ~DS, Table 1). This supports the idea that deploying Autoparty across the full docking results may effectively uncover buried hits missed by the dock scoring function. Further, while Autoparty did not outperform standard visual inspection, it exceeded the naive hit rate of "All" tested compounds (Table 1).

Two points warrant emphasis. First, the original virtual screens and hit selection procedures, including the visual

inspection, were performed by different people than the blind retrospective Autoparty hit picking case study, likely affecting the prioritized interactions and the observed hit rates. This fact is highlighted by the results from retrospectively grading the experimentally tested compounds, in which the correlation with assigned grades and hit rates was much stronger for AmpC (Supporting Table 4). A greater understanding of the important ligand interactions for a given binding site is critical to achieve a significant improvement in hit rate when using Autoparty. Second, while Autoparty failed to improve retrospective hit rates for well-scoring D4 molecules, its predictions provided access to a larger region of chemical space than manual inspection (Figure 4f). Experimental testing from this previously inaccessible region may improve hit rates, as seen with AmpC, where Autoparty prioritized different regions of chemical space than DOCK score and improved hit rates. The full list of assigned grades and Autoparty predictions for both targets are available in the Supporting Data sets.

Prospective Case Study: Pharma Campaign. We assessed Autoparty's hit identification efficiency in a structure-based virtual screening campaign at Pfizer. The Pfizer team screened an internal library against an undisclosed protein binding site using Glide.⁶ They uploaded 4,792 compounds to Autoparty, prefiltered based on a docking score cutoff of -6.5 and a strain energy cutoff of 5 kcal/mol. The team leveraged an unpublished ligand-protein complex to prioritize binding site pharmacophore features, specifically hydrophobic and hydrogen bond interactions with the protein. Annotators assigned grade A to docked compounds satisfying the most favorable pharmacophore

features and grades from B through D to those with weaker interactions. Grade F denoted docked compounds exhibiting illogical interactions (e.g., a nonpolar group occupying a polar/charged region). A total of 72 compounds were annotated, and the predicted grades for all the compounds are illustrated in Figure 5a. Four of the 31 compounds with expert-assigned A grades were readily available and selected for testing. In addition, the team short-listed 1,521 potential compounds with both predicted A grades and high shape complementarity (hereon referred to as “Other Methods”, see Supporting Methods), again selecting a subset of 193 molecules through clustering for experimental testing. Of these 193 molecules, 20 compounds exhibited IC₅₀ values below 30 μ M for a primary hit rate of 10.4% (Figure 5b and Table 2). In comparison, the primary hit rate using only other methods (OM) was 7.4%; Autoparty by itself (AP) achieved a hit rate 40% higher.

Further, OM compounds receiving a vote from Autoparty had double the hit rate of those that lacked it (10.8% AP&OM vs 5.3% ~AP&OM, Table 1), effectively filtering out inactive compounds. Assessed at increasing potency thresholds, all OM molecules that achieved IC₅₀s below 10 μ M were likewise identified by Autoparty, as was the singular sub-1 μ M hit (Table 1). As a final point, Autoparty's utility could extend beyond virtual screening. Rather than learning from user-assigned annotations by human inspection, medicinal chemistry compounds could be annotated based on experimental values. For instance, if these data are available, a researcher might instead assign compound grades by binned IC₅₀ values.

Autoparty Implementation. We primarily wrote Autoparty's back-end service in Python to handle processing the provided input files, calculating protein–ligand intermolecular interactions, reading and writing to the SQL database, training models, generating model predictions, and selecting examples to service to the front end. See Supporting Figure 3 for the full SQL database schema. All ML models are implemented in the Pytorch deep learning library.⁵⁵ The front-end graphical user interface is implemented in JavaScript, HTML, and CSS. The molecular visualization panel uses 3Dmol.js for on-page structure viewing and displaying calculated interactions.⁵⁶ The Celery queue⁵⁷ and Redis broker⁵⁸ handle asynchronous tasks such as updating the database, training models, and creating output prediction summaries. We containerized the application for portability using Singularity (now Apptainer).^{59,60}

■ ASSOCIATED CONTENT

Data Availability Statement

The Autoparty source code, example input data files, and containerized application are all available at <https://github.com/keiserlab/autoparty>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.5c00850>.

Supporting Data sets 1–4 on results from D4/AmpC virtual screens with assigned and predicted grades, experimentally tested molecules with associated binding affinities, grades, and Autoparty predictions (XLSX)

Uncertainty estimation, nominal vs ordinal label encoding strategies, database schema (Supporting Figures 1–3); Autoparty platform and LUNA library settings, uncertainty quantification testing, blinded retrospective grading; and methods for model training, uncertainty

quantification, metrics, and case study protocols (Supporting Tables 1–4) (PDF)

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Author Contributions

L.S. designed and wrote the Autoparty software with input from M.K., F.L., A.M.M., and M.J.K. F.L. and A.M.M. provided testing and feedback on its implementation. D.F.M. performed initial uncertainty quantification studies. B.W.H. performed retrospective docking studies and analyses. L.S. wrote the manuscript with input from M.K., D.F.M., B.W.H., F.L., and M.J.K. M.J.K. and A.M.M. supervised the project.

Notes

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REFERENCES

- (1) Bharatam, P. V. Computer-Aided Drug Design. In *Drug Discovery and Development: From Targets and Molecules to Medicines*; Poduri, R., Ed.; Springer, 2021; pp 137–210.
- (2) Kontoyianni, M. Docking and Virtual Screening in Drug Discovery. In *Proteomics for Drug Discovery: Methods and Protocols*; Lazar, I. M.; Kontoyianni, M.; Lazar, A. C., Eds.; Springer, 2017; pp 255–266.
- (3) Lionta, E.; Spyrou, G.; Vassilatis, D. K.; Cournia, Z. Structure-Based Virtual Screening for Drug Discovery: Principles, Applications and Recent Advances. *Curr. Top. Med. Chem.* **2014**, *14* (16), 1923–1938.
- (4) Coleman, R. G.; Carchia, M.; Sterling, T.; Irwin, J. J.; Shoichet, B. K. Ligand Pose and Orientational Sampling in Molecular Docking. *PLoS One* **2013**, *8* (10), No. e75992.
- (5) Irwin, J. J.; Shoichet, B. K. Docking Screens for Novel Ligands Conferring New Biology. *J. Med. Chem.* **2016**, *59* (9), 4103–4120.
- (6) Friesner, R. A.; Banks, J. L.; Murphy, R. B.; Halgren, T. A.; Klicic, J. J.; Mainz, D. T.; Repasky, M. P.; Knoll, E. H.; Shelley, M.; Perry, J. K.; Shaw, D. E.; Francis, P.; Shenkin, P. S. Glide: A New Approach for Rapid, Accurate Docking and Scoring. 1. Method and Assessment of Docking Accuracy. *J. Med. Chem.* **2004**, *47* (7), 1739–1749.
- (7) Ban, T.; Ohue, M.; Akiyama, Y. Multiple grid arrangement improves ligand docking with unknown binding sites: Application to the inverse docking problem. *Comput. Biol. Chem.* **2018**, *73*, 139–146, DOI: 10.1016/j.compbiolchem.2018.02.008.
- (8) Maia, E. H. B.; Assis, L. C.; de Oliveira, T. A.; da Silva, A. M.; Taranto, A. G. Structure-Based Virtual Screening: From Classical to Artificial Intelligence. *Front. Chem.* **2020**, *8*, No. 343, DOI: 10.3389/fchem.2020.00343.
- (9) Fischer, A.; Smiško, M.; Sellner, M.; Lill, M. A. Decision Making in Structure-Based Drug Discovery: Visual Inspection of Docking Results. *J. Med. Chem.* **2021**, *64* (5), 2489–2500.
- (10) Xu, M.; Shen, C.; Yang, J.; Wang, Q.; Huang, N. Systematic Investigation of Docking Failures in Large-Scale Structure-Based Virtual Screening. *ACS Omega* **2022**, *7* (43), 39417–39428.
- (11) Wu, Y.; Liu, F.; Glenn, I.; Fonseca-Valencia, K.; Paris, L.; Xiong, Y.; Jerome, S. V.; Brooks, C. L. I.; Shoichet, B. K. Identifying Artifacts from Large Library Docking. *J. Med. Chem.* **2024**, *67* (18), 16796–16806.
- (12) Smith, S. T.; Meiler, J. Assessing multiple score functions in Rosetta for drug discovery. *PLoS One* **2020**, *15* (10), No. e0240450.
- (13) Chen, W.; Cui, D.; Jerome, S. V.; Michino, M.; Lenselink, E. B.; Huggins, D. J.; Beutrait, A.; Vendome, J.; Abel, R.; Friesner, R. A.; Wang, L. Enhancing Hit Discovery in Virtual Screening through Absolute Protein–Ligand Binding Free-Energy Calculations. *J. Chem. Inf. Model.* **2023**, *63* (10), 3171–3185.
- (14) Michino, M.; Beutrait, A.; Boyles, N. A.; Nadupalli, A.; Dementiev, A.; Sun, S.; Ginn, J.; Baxt, L.; Suto, R.; Bryk, R.; Jerome, S. V.; Huggins, D. J.; Vendome, J. Shape-Based Virtual Screening of a Billion-Compound Library Identifies Mycobacterial Lipamide Dehydrogenase Inhibitors. *ACS Bio Med. Chem. Au* **2023**, *3* (6), 507–515.
- (15) Valsson, Í.; Warren, M. T.; Deane, C. M.; Magarkar, A.; Morris, G. M.; Biggin, P. C. Narrowing the gap between machine learning scoring functions and free energy perturbation using augmented data. *Commun. Chem.* **2025**, *8* (1), No. 41.
- (16) Ripphausen, P.; Stumpfe, D.; Bajorath, J. Analysis of structure-based virtual screening studies and characterization of identified active compounds. *Future Med. Chem.* **2012**, *4* (5), 603–613.
- (17) Lyu, J.; Wang, S.; Balus, T. E.; Singh, I.; Levit, A.; Moroz, Y. S.; O'Meara, M. J.; Che, T.; Alga, E.; Tolmachova, K.; Tolmachev, A. A.; Shoichet, B. K.; Roth, B. L.; Irwin, J. J. Ultra-large library docking for discovering new chemotypes. *Nature* **2019**, *566* (7743), 224–229.
- (18) Singh, I.; Li, F.; Fink, E. A.; Chau, I.; Li, A.; Rodriguez-Hernández, A.; Glenn, I.; Zapatero-Belinchón, F. J.; Rodriguez, M. L.; Devkota, K.; Deng, Z.; White, K.; Wan, X.; Tolmachova, N. A.; Moroz, Y. S.; Kaniskan, H. Ü.; Ott, M.; García-Sastre, A.; Jin, J.; Shoichet, B. K.; et al. Structure-Based Discovery of Inhibitors of the SARS-CoV-2 Nsp14 N7-Methyltransferase. *J. Med. Chem.* **2023**, *66* (12), 7785–7803.
- (19) Fink, E. A.; Bardine, C.; Gahbauer, S.; Singh, I.; Detomasi, T. C.; White, K.; Gu, S.; Wan, X.; Chen, J.; Ary, B.; Glenn, I.; O'Connell, J.; O'Donnell, H.; Fajtová, P.; Lyu, J.; Vigneron, S.; Young, N. J.; Kondratov, I. S.; Alisoltani, A.; Craik, C. S.; et al. Large library docking for novel SARS-CoV-2 main protease non-covalent and covalent inhibitors. *Protein Sci.* **2023**, *32* (8), No. e4712.
- (20) Singh, I.; Seth, A.; Billesbølle, C. B.; Braz, J.; Rodriguiz, R. M.; Roy, K.; Bekele, B.; Craik, V.; Huang, X.-P.; Boytsov, D.; Pogorelov, V. M.; Lak, P.; O'Donnell, H.; Sandtner, W.; Irwin, J. J.; Roth, B. L.; Basbaum, A. I.; Wetsel, W. C.; Manglik, A.; Shoichet, B. K.; Rudnick, G. Structure-based discovery of conformationally selective inhibitors of the serotonin transporter. *Cell* **2023**, *186* (10), 2160–2175.e17.
- (21) Kaplan, A. L.; Confair, D. N.; Kim, K.; Barros-Álvarez, X.; Rodriguiz, R. M.; Yang, Y.; Kweon, O. S.; Che, T.; McCorry, J.; Kamber, D. N.; Phelan, J. P.; Martins, L. C.; Pogorelov, V. M.; DiBerto, J. F.; Slocum, S. T.; Huang, X.-P.; Kumar, J. M.; Robertson, M. J.; Panova, O.; Ellman, J. A.; et al. Bespoke library docking for 5-HT_{2A} receptor agonists with anti-depressant activity. *Nature* **2022**, *610* (7932), 582–591.
- (22) Levit Kaplan, A.; Strachan, R. T.; Braz, J. M.; Craik, V.; Slocum, S.; Mangano, T.; Amabo, V.; O'Donnell, H.; Lak, P.; Basbaum, A. I.; Roth, B. L.; Shoichet, B. K. Structure-Based Design of a Chemical Probe Set for the 5-HT_{2A} Serotonin Receptor. *J. Med. Chem.* **2022**, *65* (5), 4201–4217.
- (23) Gahbauer, S.; Correy, G. J.; Schuller, M.; Ferla, M. P.; Doruk, Y. U.; Rachman, M.; Wu, T.; Diolaiti, M.; Wang, S.; Neitz, R. J.; Fearon, D.; Radchenko, D. S.; Moroz, Y. S.; Irwin, J. J.; Renslo, A. R.; Taylor, J. C.; Gestwicki, J. E.; Von Delft, F.; Ashworth, A.; Fraser, J. S.; et al. Iterative computational design and crystallographic screening identifies potent inhibitors targeting the Nsp3 macrodomain of SARS-CoV-2. *Proc. Natl. Acad. Sci. U.S.A.* **2023**, *120* (2), No. e2212931120.
- (24) Liu, F.; Mailhot, O.; Glenn, I. S.; Vigneron, S. F.; Bassim, V.; Xu, X.; Fonseca-Valencia, K.; Smith, M. S.; Radchenko, D. S.; Fraser, J. S.; Moroz, Y. S.; Irwin, J. J.; Shoichet, B. K. The impact of library size and scale of testing on virtual screening. *Nat. Chem. Biol.* **2025**, *21*, 1039–1045.
- (25) Fink, E. A.; Xu, J.; Hübner, H.; Braz, J. M.; Seemann, P.; Avet, C.; Craik, V.; Weikert, D.; Schmidt, M. F.; Webb, C. M.; Tolmachova, N. A.; Moroz, Y. S.; Huang, X.-P.; Kalyanaraman, C.; Gahbauer, S.; Chen, G.; Liu, Z.; Jacobson, M. P.; Irwin, J. J.; Gmeiner, P.; et al. Structure-based discovery of nonopioid analgesics acting through the α 2A-adrenergic receptor. *Science* **2022**, *377* (6614), No. eabn7065.
- (26) Alon, A.; Lyu, J.; Braz, J. M.; Tummino, T. A.; Craik, V.; O'Meara, M. J.; Webb, C. M.; Radchenko, D. S.; Moroz, Y. S.; Huang, X.-P.; Liu, Y.; Roth, B. L.; Irwin, J. J.; Basbaum, A. I.; Shoichet, B. K.; Kruse, A. C. Structures of the σ 2 receptor enable docking for bioactive ligand discovery. *Nature* **2021**, *600* (7890), 759–764.
- (27) Schulz-Gasch, T.; Stahl, M. Scoring functions for protein–ligand interactions: A critical perspective. *Drug Discovery Today: Technol.* **2004**, *1* (3), 231–239.
- (28) Ruddigkeit, L.; van Deursen, R.; Blum, L. C.; Reymond, J.-L. Enumeration of 166 Billion Organic Small Molecules in the Chemical Universe Database GDB-17. *J. Chem. Inf. Model.* **2012**, *52* (11), 2864–2875.
- (29) Lyu, J.; Irwin, J. J.; Shoichet, B. K. Modeling the expansion of virtual screening libraries. *Nat. Chem. Biol.* **2023**, *19* (6), 712–718.
- (30) Choung, O.-H.; Vianello, R.; Segler, M.; Stieff, N.; Jiménez-Luna, J. Extracting medicinal chemistry intuition via preference machine learning. *Nat. Commun.* **2023**, *14* (1), No. 6651.
- (31) Gomez, L. Decision Making in Medicinal Chemistry: The Power of Our Intuition. *ACS Med. Chem. Lett.* **2018**, *9* (10), 956–958.

- (32) Ren, P.; Xiao, Y.; Chang, X.; Huang, P.-Y.; Li, Z.; Gupta, B. B.; Chen, X.; Wang, X. A Survey of Deep Active Learning. *ACM Comput. Surv.* **2021**, *54* (9), 180:1–180:40.
- (33) Mosqueira-Rey, E.; Hernández-Pereira, E.; Alonso-Ríos, D.; Bobes-Bascarán, J.; Fernández-Leal, Á. Human-in-the-loop machine learning: A state of the art. *Artif. Intell. Rev.* **2023**, *56* (4), 3005–3054.
- (34) Yang, Y.; Yao, K.; Repasky, M. P.; Leswing, K.; Abel, R.; Shoichet, B. K.; Jerome, S. V. Efficient Exploration of Chemical Space with Docking and Deep Learning. *J. Chem. Theory Comput.* **2021**, *17* (11), 7106–7119.
- (35) Graff, D. E.; Shakhnovich, E. I.; Coley, C. W. Accelerating high-throughput virtual screening through molecular pool-based active learning. *Chem. Sci.* **2021**, *12* (22), 7866–7881.
- (36) Thompson, J.; Walters, W. P.; Feng, J. A.; Pabon, N. A.; Xu, H.; Maser, M.; Goldman, B. B.; Moustakas, D.; Schmidt, M.; York, F. Optimizing active learning for free energy calculations. *Artif. Intell. Life Sci. Sci.* **2022**, *2*, No. 100050.
- (37) Fassio, A. V.; Shub, L.; Ponzoni, L.; McKinley, J.; O'Meara, M. J.; Ferreira, R. S.; Keiser, M. J.; de Melo Minardi, R. C. Prioritizing Virtual Screening with Interpretable Interaction Fingerprints. *J. Chem. Inf. Model.* **2022**, *62* (18), 4300–4318.
- (38) O'Boyle, N. M.; Banck, M.; James, C. A.; Morley, C.; Vandermeersch, T.; Hutchison, G. R. Open Babel: An open chemical toolbox. *J. Cheminf.* **2011**, *3* (1), No. 33.
- (39) RDKit, <https://www.rdkit.org/>. (Retrieved April 22, 2024).
- (40) Wójcikowski, M.; Kukiela, M.; Stepniewska-Dziubinska, M. M.; Siedlecki, P. Development of a protein-ligand extended connectivity (PLEC) fingerprint and its application for binding affinity predictions. *Bioinformatics* **2019**, *35* (8), 1334–1341.
- (41) van Tilborg, D.; Alenicheva, A.; Grisoni, F. Exposing the Limitations of Molecular Machine Learning with Activity Cliffs. *J. Chem. Inf. Model.* **2022**, *62* (23), S938–S951.
- (42) Rogers, D.; Hahn, M. Extended-Connectivity Fingerprints. *J. Chem. Inf. Model.* **2010**, *50* (5), 742–754.
- (43) Scalia, G.; Grambow, C. A.; Pernici, B.; Li, Y.-P.; Green, W. H. Evaluating Scalable Uncertainty Estimation Methods for DNN-Based Molecular Property Prediction. 2019, arXiv:1910.03127. arXiv.org e-Printarchive. <http://arxiv.org/abs/1910.03127>.
- (44) Hirschfeld, L.; Swanson, K.; Yang, K.; Barzilay, R.; Coley, C. W. Uncertainty Quantification Using Neural Networks for Molecular Property Prediction. *J. Chem. Inf. Model.* **2020**, *60* (8), 3770–3780.
- (45) Lakshminarayanan, B.; Pritzel, A.; Blundell, C. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. 2017, arXiv:1612.01474. arXiv.org e-Printarchive. <http://arxiv.org/abs/1612.01474>.
- (46) Gal, Y.; Ghahramani, Z. In *Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning*; Proceedings of The 33rd International Conference on Machine Learning; PMLR, 2016; pp 1050–1059.
- (47) Berenger, F.; Yamanishi, Y. A Distance-Based Boolean Applicability Domain for Classification of High Throughput Screening Data. *J. Chem. Inf. Model.* **2019**, *59* (1), 463–476.
- (48) Liu, R.; Wallqvist, A. Molecular Similarity-Based Domain Applicability Metric Efficiently Identifies Out-of-Domain Compounds. *J. Chem. Inf. Model.* **2019**, *59* (1), 181–189.
- (49) Settles, B.; Craven, M. In *An Analysis of Active Learning Strategies for Sequence Labeling Tasks*, Proceedings of the Conference on Empirical Methods in Natural Language Processing—EMNLP '08; ACM, 2008; pp 1070–1079.
- (50) Sharma, M.; Bilgic, M. Evidence-based uncertainty sampling for active learning. *Data Min. Knowl. Discovery* **2017**, *31* (1), 164–202.
- (51) Gusev, F.; Gutkin, E.; Kurnikova, M. G.; Isayev, O. Active learning guided drug design lead optimization based on relative binding free energy modeling. *J. Chem. Inf. Model.* **2023**, *63*, 583–594, DOI: 10.1021/acs.jcim.2c01052.
- (52) Thompson, J.; Walters, W. P.; Feng, J. A.; Pabon, N. A.; Xu, H.; Maser, M.; Goldman, B. B.; Moustakas, D.; Schmidt, M.; York, F. Optimizing active learning for free energy calculations. *Artif. Intell. Life Sci. Sci.* **2022**, *2*, No. 100050.
- (53) Larson, E. T.; Ojo, K. K.; Murphy, R. C.; Johnson, S. M.; Zhang, Z.; Kim, J. E.; Leibly, D. J.; Fox, A. M. W.; Reid, M. C.; Dale, E. J.; Perera, B. G. K.; Kim, J.; Hewitt, S. N.; Hol, W. G. J.; Verlinde, C. L. M. J.; Fan, E.; Van Voorhis, W. C.; Maly, D. J.; Merritt, E. A. Multiple Determinants for Selective Inhibition of Apicomplexan Calcium-Dependent Protein Kinase CDPK1. *J. Med. Chem.* **2012**, *55* (6), 2803–2810.
- (54) Hall, B. W.; Tummino, T. A.; Tang, K.; Irwin, J. J.; Shoichet, B. K. A database for large-scale docking and experimental results *bioRxiv* 2025.
- (55) Paszke, A.; Gross, S.; Massa, F.; Lerer, A.; Bradbury, J.; Chanan, G.; Killeen, T.; Lin, Z.; Gimelshein, N.; Antiga, L.; Desmaison, A.; Köpf, A.; Yang, E.; DeVito, Z.; Raison, M.; Tejani, A.; Chilamkurthy, S.; Steiner, B.; Fang, L.; Chintala, S. PyTorch: An Imperative Style, High-Performance Deep Learning Library. 2019, arXiv:1912.01703. arXiv.org e-Printarchive. <http://arxiv.org/abs/1912.01703>.
- (56) Rego, N.; Koes, D. 3Dmol.js: Molecular visualization with WebGL. *Bioinformatics* **2015**, *31* (8), 1322–1324.
- (57) celery/celery: Distributed Task Queue (development branch). <https://github.com/celery/celery>. (Retrieved April 22, 2024).
- (58) Redis/redis. *Redis*, 2024. <https://github.com/redis/redis>. (Original work published 2009).
- (59) Kurtzer, G. M.; cclerget; Bauer, M.; Kaneshiro, I.; Trudgian, D.; Godlove, D. *hpcng/singularity: Singularity 3.7.3 (Version v3.7.3)*, Zenodo 2021.
- (60) Apptainer/singularity: Singularity has been renamed to Apptainer as part of us moving the project to the Linux Foundation. This repo has been persisted as a snapshot right before the changes. 2024, <https://github.com/apptainer/singularity>. (Retrieved May 22, 2024).